

Incident Investigation Report

Password Spraying Attack Analysis
MITRE ATT&CK: T1110.003

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1.0 EXECUTIVE SUMMARY

This report presents the investigation of a simulated password spraying attack conducted as part of the CyberDefenders T1110.003 challenge. The investigation was performed using Splunk to analyze Windows Security Event Logs and identify malicious authentication activity.

Analysis of Event IDs 4625 and 4624 revealed that a single source IP address repeatedly attempted authentication against multiple user accounts within a short time frame, consistent with the MITRE ATT&CK Password Spraying technique (T1110.003). The attack lasted approximately 5 minutes and 48 seconds, resulting in the successful compromise of six user accounts through the Remote Desktop Protocol (RDP). Approximately 11 minutes after the password spraying activity ended, the attacker successfully authenticated to the target system.

The investigation reconstructed the attack timeline, identified key indicators of compromise (IOCs), and documented the techniques observed during the intrusion. Ultimately, the investigation successfully identified the attack source, affected accounts, and the specific attack methodology. Based on these findings, recommendations were developed to strengthen authentication security and improve the detection of password spraying attacks.

2.0 INCIDENT OVERVIEW

Field	Value
Incident Title	Password Spraying Attack Investigation
Incident Type	Password Spraying
MITRE ATT&CK	T1110.003 – Password Spraying
Severity	Medium
Investigation Date	July 2026
Analyst	Renad Gadah
Platform	
Analysis Tool	CyberDefenders
Log Source	Splunk
Attack Source IP	Windows Security Event Logs
Target Protocol	192.168.1.60
Affected Accounts	Remote Desktop Protocol (RDP)
Case ID	6 User Accounts
	IR-2026-001

The investigation focused on analyzing Windows Security Event Logs to identify the source, technique, and impact of a simulated password spraying attack. Authentication events were correlated to reconstruct the attack timeline, identify compromised accounts, and determine the protocol used to gain unauthorized access.

3.0 Investigation Scope

The scope of this investigation was limited to the analysis of Windows Security Event Logs generated within the CyberDefenders simulated environment. The objective was to identify and investigate authentication events related to a suspected password spraying attack, determine the attack source, reconstruct the sequence of events, identify compromised accounts, and map the observed activity to the MITRE ATT&CK framework.

3.1 Included in Scope

- Windows Security Event Logs
- Authentication events (Event IDs 4624 & 4625)
- Logon Type analysis
- Source IP identification
- RDP authentication activity
- Attack timeline reconstruction
- Indicators of Compromise (IOCs)
- MITRE ATT&CK technique mapping

3.2 Out of Scope

- Endpoint forensic analysis
- Malware analysis
- Memory analysis
- Network packet analysis (PCAP)
- Active incident response actions

4.0 Investigation Methodology

The investigation followed a structured log analysis approach using Splunk to examine Windows Security Event Logs. Authentication events were reviewed and correlated to identify suspicious activity, reconstruct the attack timeline, and determine the attack technique, affected accounts, and protocol used during the intrusion.

4.1 Investigation Steps

1. Reviewed failed authentication events (Event ID 4625) to identify abnormal login failures.
2. Correlated failed and successful logon events (Event IDs 4625 and 4624) to determine whether any authentication attempts were successful.
3. Analyzed logon types and source IP addresses to identify the attack vector and targeted protocol.
4. Reconstructed the sequence of events to establish the attack timeline.
5. Identified indicators of compromise (IOCs) and mapped the observed activity to the MITRE ATT&CK framework.
6. Documented findings and developed security recommendations based on the investigation.

5.0 Evidence & Analysis

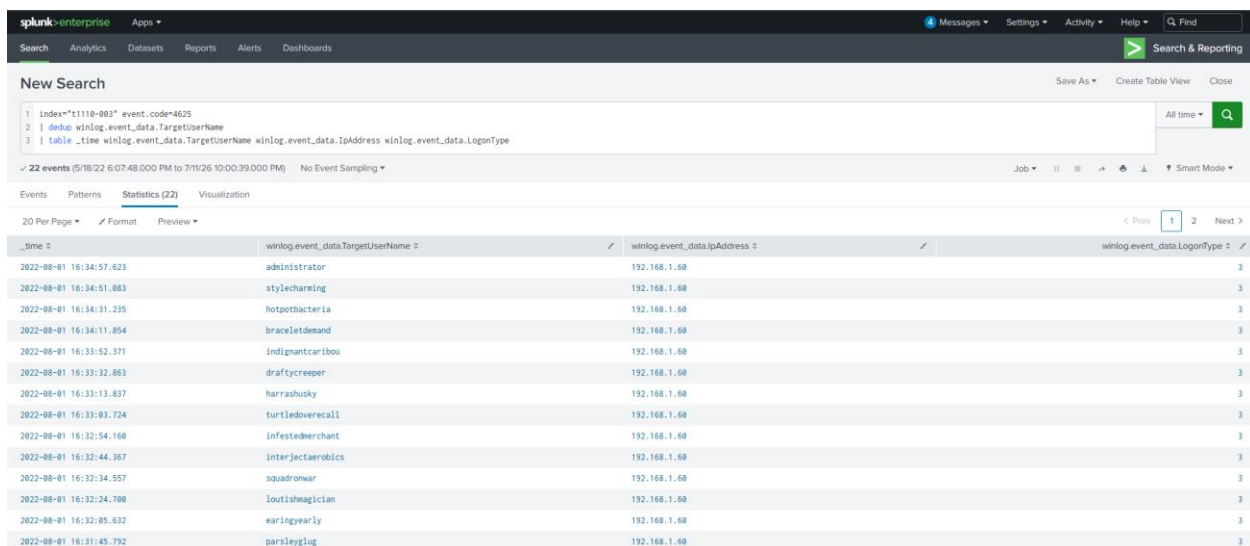
5.1 Review Failed Authentication Events

Objective:

Identify abnormal failed authentication attempts that may indicate a password spraying attack.

SPL Query:

```
index="t1110-003" event.code=4625
| dedup winlog.event_data.TargetUserName
| table _time winlog.event_data.TargetUserName winlog.event_data.IpAddress
winlog.event_data.LogonType
```



The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk-enterprise' and various menu items like Search, Analytics, Datasets, Reports, Alerts, and Dashboards. Below this, a 'New Search' panel displays the SPL query: `index="t1110-003" event.code=4625 | dedup winlog.event_data.TargetUserName | table _time winlog.event_data.TargetUserName winlog.event_data.IpAddress winlog.event_data.LogonType`. The search results are shown in a table with 22 events. The table has three columns: `_time`, `winlog.event_data.TargetUserName`, and `winlog.event_data.IpAddress`. The results show multiple failed authentication attempts from the same IP address (192.168.1.60) targeting different user accounts.

_time	winlog.event_data.TargetUserName	winlog.event_data.IpAddress	winlog.event_data.LogonType
2022-08-01 16:34:57.623	administrator	192.168.1.60	3
2022-08-01 16:34:51.083	stylechasing	192.168.1.60	3
2022-08-01 16:34:31.235	hotpotbacteria	192.168.1.60	3
2022-08-01 16:34:11.854	braceletdemand	192.168.1.60	3
2022-08-01 16:33:52.371	indignantcaribou	192.168.1.60	3
2022-08-01 16:33:32.863	draftycreeper	192.168.1.60	3
2022-08-01 16:33:13.837	harrashusky	192.168.1.60	3
2022-08-01 16:33:03.724	turtledovercall	192.168.1.60	3
2022-08-01 16:32:54.160	infestedmerchant	192.168.1.60	3
2022-08-01 16:32:44.367	interjectaerobics	192.168.1.60	3
2022-08-01 16:32:34.557	squadronar	192.168.1.60	3
2022-08-01 16:32:24.700	loutishmagician	192.168.1.60	3
2022-08-01 16:32:05.632	earinyearly	192.168.1.60	3
2022-08-01 16:31:45.792	parsleyglug	192.168.1.60	3

Figure 5.1: Failed authentication events returned by the Splunk query.

Analysis:

Multiple failed authentication attempts targeting different user accounts were observed from the same source IP address, indicating password spraying behavior.

No evidence indicated repeated attempts against a single account, further supporting password spraying rather than traditional brute-force activity.

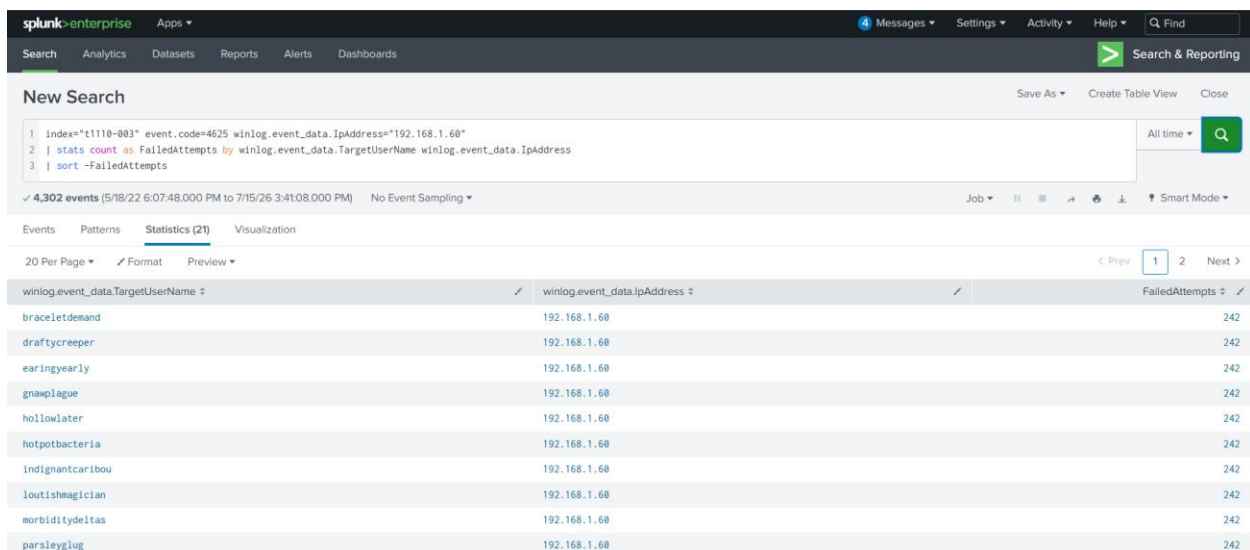
5.2 Analyze Password Spraying Pattern

Objective:

Analyze failed authentication events to determine whether the observed activity matches the characteristics of a password spraying attack.

SPL Query:

```
index="t1110-003" event.code=4625
winlog.event_data.IpAddress="192.168.1.60"
| stats count as FailedAttempts by winlog.event_data.TargetUserName
winlog.event_data.IpAddress
| sort -FailedAttempts
```



The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk enterprise' logo and various menu items like Messages, Settings, Activity, Help, and a search bar. Below this is a 'New Search' section with a search bar containing the SPL query. The search results are displayed in a table with columns for 'winlog.event_data.TargetUserName', 'winlog.event_data.IpAddress', and 'FailedAttempts'. The table shows 10 rows of data, all with the same IP address '192.168.1.60' and varying user names, each with a count of 242 failed attempts.

winlog.event_data.TargetUserName	winlog.event_data.IpAddress	FailedAttempts
braceletdemand	192.168.1.60	242
draftycreeper	192.168.1.60	242
earinyearly	192.168.1.60	242
gnawplague	192.168.1.60	242
hollowlater	192.168.1.60	242
hotpotbacteria	192.168.1.60	242
indignantcaribou	192.168.1.60	242
loutishmagician	192.168.1.60	242
morbiditydeltas	192.168.1.60	242
parsleyglug	192.168.1.60	242

Figure 5.2: Multiple failed authentication attempts from a single source IP targeting different user accounts.

Analysis:

The failed authentication attempts were distributed across multiple user accounts rather than concentrated on a single account. This distribution is a key characteristic of password spraying attacks, where attackers attempt a common password against many accounts to reduce the likelihood of triggering account lockout policies.

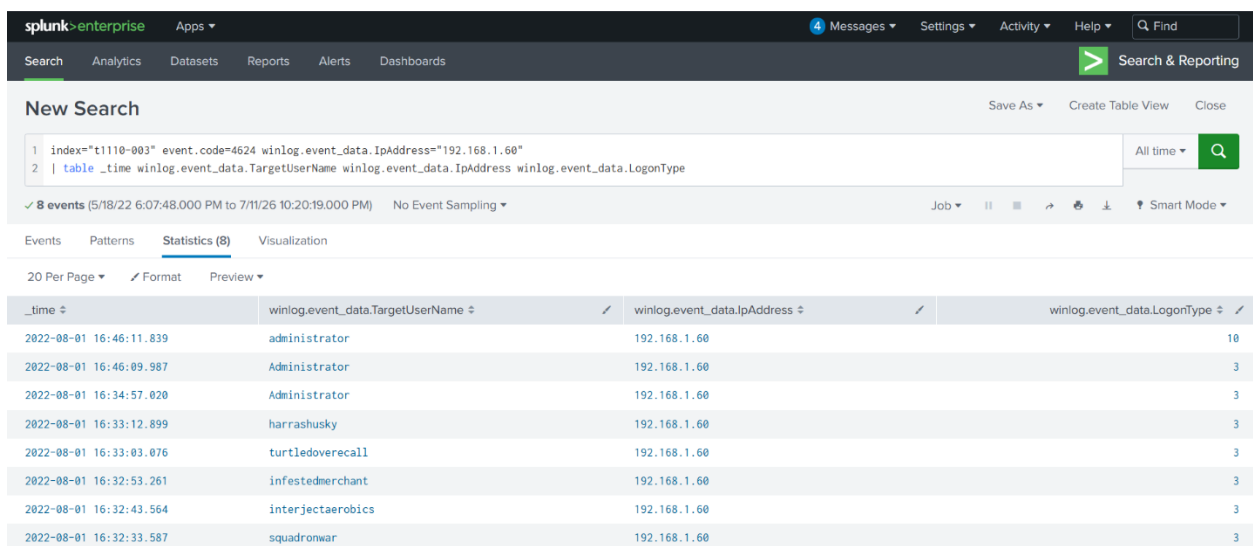
5.3 Identify Successful Authentication

Objective:

Determine whether the password spraying attack resulted in successful authentication and identify the compromised accounts.

SPL Query:

```
index="t1110-003" event.code=4624
winlog.event_data.IpAddress="192.168.1.60"
| table _time winlog.event_data.TargetUserName winlog.event_data.IpAddress
winlog.event_data.LogonType
```



The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Messages', 'Settings', 'Activity', and 'Help'. Below this is a 'Search & Reporting' section. The main area displays a 'New Search' result with a table of 8 events. The table has columns for '_time', 'winlog.event_data.TargetUserName', 'winlog.event_data.IpAddress', and 'winlog.event_data.LogonType'. The data shows several successful logins for different users from the IP 192.168.1.60.

_time	winlog.event_data.TargetUserName	winlog.event_data.IpAddress	winlog.event_data.LogonType
2022-08-01 16:46:11.839	administrator	192.168.1.60	10
2022-08-01 16:46:09.987	Administrator	192.168.1.60	3
2022-08-01 16:34:57.020	Administrator	192.168.1.60	3
2022-08-01 16:33:12.899	harrashusky	192.168.1.60	3
2022-08-01 16:33:03.076	turtledoverecall	192.168.1.60	3
2022-08-01 16:32:53.261	infestedmerchant	192.168.1.60	3
2022-08-01 16:32:43.564	interjectaerobics	192.168.1.60	3
2022-08-01 16:32:33.587	squadronwar	192.168.1.60	3

Figure 5.3: Successful authentication events identified after the password spraying attack.

Analysis:

The analysis identified successful authentication events following the password spraying activity. A total of six user accounts were successfully authenticated, confirming that the attacker was able to compromise multiple accounts after repeated failed login attempts. This indicates that the password spraying attack was successful.

5.4 Determine the Targeted Protocol

Objective:

Identify the protocol used by the attacker to gain unauthorized access to the target system.

SPL Query:

```
index="t1110-003" event.code=261
```

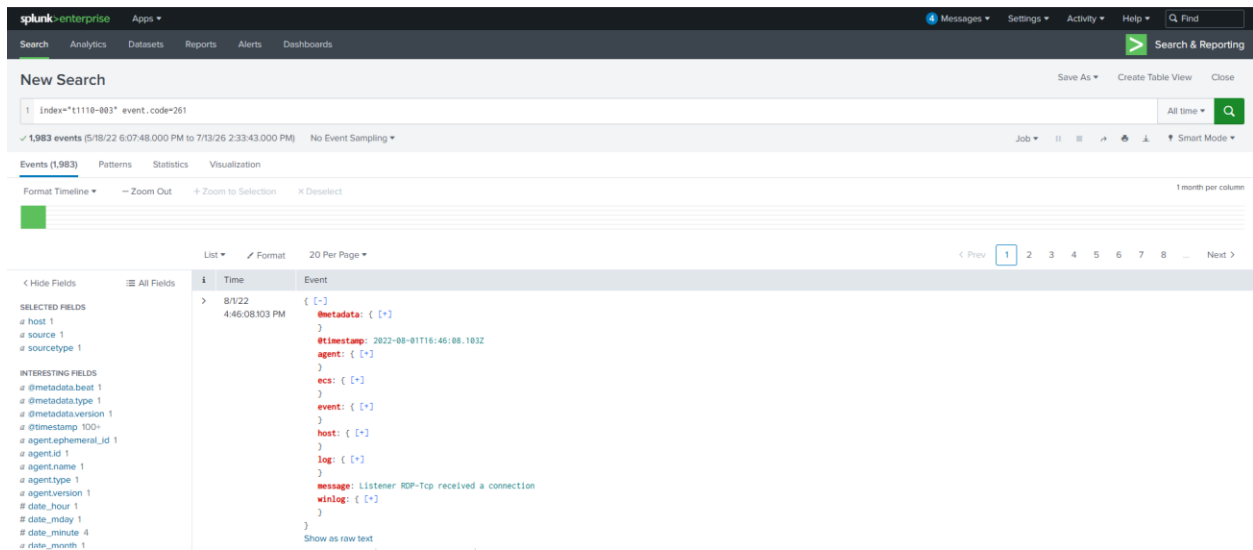


Figure 5.4: Evidence confirming the use of the Remote Desktop Protocol (RDP).

Analysis:

The investigation identified the event "Listener RDP-Tcp received a connection," confirming that the attacker established a Remote Desktop Protocol (RDP) session following successful authentication. This indicates that RDP was the protocol used to gain interactive access to the target system.

5.5 Reconstruct the Attack Timeline

Objective:

Reconstruct the sequence of events to understand the progression of the password spraying attack from the initial authentication attempts to the successful system access.

SPL Query:

```
index="t1110-003"
winlog.event_data.IpAddress="192.168.1.60"
(event.code=4625 OR event.code=4624)
| sort _time
| streamstats count
| eventstats max(count) as total
| where count<=5 OR count>=total-4
| table _time event.code winlog.event_data.TargetUserName
winlog.event_data.LogonType
```

The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk-enterprise' and 'Apps'. Below it, a search bar contains the query: `index="t1110-003" winlog.event_data.IpAddress="192.168.1.60" (event.code=4625 OR event.code=4624) | sort _time | streamstats count | eventstats max(count) as total | where count<=5 OR count>=total-4 | table _time event.code winlog.event_data.TargetUserName winlog.event_data.LogonType`. The results are displayed in a table with columns: `_time`, `event.code`, `winlog.event_data.TargetUserName`, and `winlog.event_data.LogonType`. The table shows 10 events, with the first 9 having a count of 3 and the last one having a count of 10.

_time	event.code	winlog.event_data.TargetUserName	winlog.event_data.LogonType
2022-08-01 16:29:09.460	4625	yokehorde	3
2022-08-01 16:29:09.465	4625	yokehorde	3
2022-08-01 16:29:09.617	4625	yokehorde	3
2022-08-01 16:29:09.617	4625	yokehorde	3
2022-08-01 16:29:09.804	4625	yokehorde	3
2022-08-01 16:34:57.592	4625	administrator	3
2022-08-01 16:34:57.613	4625	administrator	3
2022-08-01 16:34:57.623	4625	administrator	3
2022-08-01 16:45:09.987	4624	Administrator	3
2022-08-01 16:46:11.839	4624	administrator	10

Figure 5.5: Chronological sequence of authentication events associated with the password spraying attack.

Time	Event ID	Description
16:31:26	4625	First failed authentication attempt detected
16:37:14	4625	Password spraying activity concluded
16:48:09	4624	Successful authentication achieved
16:48:11	261	RDP connection established

Analysis:

The chronological analysis shows that the attack began with multiple failed authentication attempts originating from the same source IP address (192.168.1.60). These attempts targeted multiple user accounts over a period of approximately **5 minutes and 48 seconds**, indicating password spraying activity. Approximately **11 minutes** after the attack concluded, the attacker successfully authenticated and established remote access to the target system via RDP. This sequence confirms the progression of the attack from credential guessing to successful compromise.

6.0 Key Findings

- The attack was identified as a Password Spraying attack (MITRE ATT&CK T1110.003).
- The attacker originated from 192.168.1.60.
- Six user accounts were successfully compromised.
- The password spraying activity lasted approximately 5 minutes and 48 seconds.
- The attacker re-authenticated approximately 11 minutes after the attack.
- RDP was used to access the target system.

7.0 Indicators of Compromise (IOCs)

IOC Type	Value
Source IP	192.168.1.60
Event ID	4625
Event ID	4624
Protocol	RDP
MITRE Technique	T1110.003

8.0 MITRE ATT&CK Mapping

Tactic	Technique	ID
Credential Access	Password Spraying	T1110.003
Lateral Movement	Remote Services (RDP)	T1021.001

9.0 Recommendations

- Enable Multi-Factor Authentication (MFA).
- Enforce strong password policies.
- Configure Account Lockout Policies.
- Monitor repeated failed authentication attempts.
- Restrict Remote Desktop Protocol (RDP) access.
- Generate SIEM alerts for password spraying behavior.

10.0 Conclusion

The investigation confirmed that the observed activity was a successful password spraying attack. Analysis of Windows Security Event Logs enabled the identification of the attack source, compromised accounts, authentication method, and attack timeline. The findings highlight the importance of continuous log monitoring, strong authentication controls, and proactive detection mechanisms to reduce the risk of credential-based attacks.

11.0 References

- Microsoft Windows Security Auditing Documentation
- MITRE ATT&CK Framework
- CyberDefenders T1110.003 Challenge

12.0 Disclaimer

This report was prepared for educational and portfolio purposes based on a simulated CyberDefenders incident. The investigation reflects analysis performed within a controlled training environment and does not represent a real-world production incident.